

UAV Flight Analyzer

Why This Tool Exists and Why It Matters

Version 1.0 | April 2026
ArduPilot DataFlash .BIN
Quadcopter • VTOL • Fixed-Wing

After every UAV mission — how do you know the aircraft is safe to fly again?

ArduPilot flight controllers log thousands of data points every second. That raw .BIN file on the SD card contains everything needed to assess flight health — battery condition, motor balance, vibration, GPS integrity, control performance, and more. The problem is that almost nobody reads it properly, because doing so manually requires specialist tools, deep ArduPilot knowledge, and up to an hour per flight. Most teams skip it entirely and fly again on assumption. **UAV Flight Analyzer changes that.**

The Problem — Why Manual Log Review Fails

Time Cost

Thorough manual review of a single .BIN log takes 30–60 minutes using Mission Planner or MAVExplorer. Across a fleet of 10 drones flying daily, this is operationally impossible.

Expertise Barrier

Interpreting telemetry data requires deep ArduPilot knowledge. Most operators and field technicians lack the training to identify degradation patterns in raw log data.

No Standardisation

Without defined thresholds, two engineers reviewing the same log will reach different conclusions. Grounding decisions become subjective rather than data-driven.

Silent Degradation

Battery internal resistance rises gradually. Vibration increases slowly. Motor imbalance builds flight by flight. Without trend tracking, degradation is invisible until it causes a failure.

How UAV Flight Analyzer Solves It

1

Upload .BIN Log

Drop the raw DataFlash file from the flight controller SD card into the web interface.

2

18 Modules Run

Battery, motors, vibration, GPS, control, efficiency, airspeed, ESC, altitude — all checked automatically.

3

Score & Grade

Weighted health score 0–100 with A–F grade. 110+ issue codes with observed values and thresholds.

4

Recommendations

Prioritised, specific maintenance actions generated from every detected issue — not generic advice.

5

Fleet Trends

Linear regression across flight history flags rising battery IR, vibration growth, or score decline before failure.

Who Benefits

github.com/NAGASURYA444/UAV-Flight-Analyzer/issues